Module 17

PROCESS DOCUMENTATION

1. Information	
Process Name:	Supply Inventory Tracking Process
Process Owner:	Clinic Doctor / Clinic Owner
Department/Unit:	Inventory, Supplies & Operations
Document Prepared By:	Tristan Paolo C. De Castro
Date Prepared:	September 12, 2026
Version Number:	1.0
Last Updated:	September 12, 2026

2. Overview

Purpose: To replace the current reactive method — noticing supplies are low only while using them — with zero manual counting required.

Process Description: The system maintains a master list of supplies with quantity on hand. When the doctor charts a procedure and taps supplies used, the system auto-deducts them. Real-time quantity is visible on the dashboard, and full deduction history is retained.

Actors Involved:
- Doctor — taps supplies used during charting
- System — deducts, tracks, displays

3. Inputs

- Supplies Used - Module 9 (Dental Digital Charting)
- Master Supply List - Module 17 config

4. Activities

1.System maintains master supply list - System
2. Doctor taps supplies used during charting - Doctor
3. System auto-deducts supplies - System
4. System updates real-time quantity - System
5. System keeps deduction history - System
6. Doctor views live count on dashboard - Doctor

5. Outputs

- Live Stock Count -> Doctor (Module 21: Admin & Operations Dashboard)
- Deduction History -> Module 17
- Low-Stock Trigger -> Module 18 (Low-Stock Alert)

6. Stakeholders Analysis

Internal:
- Doctor / Clinic Owner — views stock, uses supplies
- IT Department — maintains inventory system

External:
- Suppliers — receive orders (via Module 20)

Note: *No receptionist is involved. Stock deducts automatically during charting.*

7. Flow Summary

Start → Doctor Charts Procedure → Tap Supplies Used → System Auto-Deducts → Update Live Count → End

8. Performance Metrics	
Stock counting	Manual, periodic (Target: Real-time)
Deduction logging	Inconsistent (Target: Automatic)
Stockout incidents	Common (Target: Rare)
Audit time	Hours (Target: Zero)

9. Issues and Challenges

- Supplies noticed as low only during use
- No real-time count
- Manual counting is periodic and error-prone
- No deduction history

10. Improvement Opportunities

- Real-time count
- Auto-deduction via charting - with Module 9 (Dental Digital Charting)
- Deduction history
- Dashboard view - with Module 21 (Admin & Operations Dashboard)

11. Automation Potential

- Deduction - Automatic
- Count update - Automatic
- History logging - Automatic

Only remaining human touchpoint: Doctor taps supplies used (one action).

12. Business Process Lifecycle Analysis	
Design	Create a master supply list with real-time deduction.
Execution	Doctor taps supplies; the system deducts automatically.
Monitoring	Track usage patterns and stockout frequency.
Analysis	Identify high-use supplies and waste.
Optimization	Adjust thresholds and reorder points.

13. Evaluation and Recommendation

Manual supply tracking is reactive and error-prone. Module 17 makes it automatic — one tap during charting keeps everything accurate.

Recommendation:

- Implement Module 17 alongside Module 9 (Dental Digital Charting**)**— they work together.
- Pre-load common supplies per procedure — reduces taps.
- Make the count visible on the dashboard — always know what's low.

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Module 18

Module 18: Low-Stock, Critical Low-Stock, and Overstock Alert Process

1. Information

Field	Details
Process Name	Low-Stock, Critical Low-Stock, and Overstock Alert Process
Process Owner	Clinic Doctor / Clinic Owner
Department/Unit	Inventory, Supplies & Operations
Document Prepared By	Tristan Paolo C. De Castro
Date Prepared	September 12, 2026
Version Number	2.1
Last Updated	September 25, 2026

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2. Overview

Purpose: To give advance, severity-based warning for **critical low stock**, **low stock**, and **overstock**, and to automatically trigger ordering for Critical Low and Low conditions — eliminating manual approval and mid-use discovery of shortages.

Process Description: The system monitors stock levels from Module 17 (Supply Inventory Tracking). It compares the live quantity against three configured thresholds per item: **Critical Low**, **Low/Reorder**, and **Overstock/Max**. When a threshold is breached, the system flags the item, notifies the doctor via the dashboard, and:

- **Critical Low** and **Low** → automatically trigger Module 20 to draft and dispatch an order (end-of-day batch, or immediate for Critical if configured).
- **Overstock** → suppress ordering and create a review note for the doctor.

Actors Involved:

- **System** – monitors, compares, classifies severity, flags, notifies, triggers Module 20
- **Doctor** – views alerts, reviews Overstock notes, handles exceptions

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Threshold and Alert Definitions

Threshold	Condition	Alert Type	Severity	Automatic Action

Critical Low	Qty ≤ Critical Low threshold	Critical Low-Stock Alert	Urgent / Red	Auto-trigger Module 20 (immediate or end-of-day batch)
Low / Reorder	Qty ≤ Reorder threshold but > Critical Low	Low-Stock Alert	Warning / Yellow	Auto-trigger Module 20 (end-of-day batch)
Overstock	Qty ≥ Overstock threshold	Overstock Alert	Informational / Blue	Suppress ordering; log review note for doctor

If multiple thresholds are breached, the highest severity wins: **Critical Low > Low > Overstock**.

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3. Inputs

- **Live Stock Quantity** – Module 17 (Supply Inventory Tracking)
- **Threshold per Item** – Module 18 config (Critical Low, Low/Reorder, Overstock/Max)
- **Pending / Queued Orders** – Module 20 (to avoid duplicate orders)
- **Expiration Dates** – Module 19 (for overstock expiry risk)
- **Usage Trends** – Module 9 (Dental Digital Charting) and Module 17

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4. Activities

1. System maintains threshold configuration per item — Critical Low, Low/Reorder, Overstock/Max — **System / Doctor**
2. System monitors live stock levels — **System**
3. System compares quantity against all configured thresholds — **System**
4. If quantity ≤ Critical Low threshold, system flags item as **Critical**, sends **Critical Low-Stock Alert**, and **auto-triggers Module 20** — **System**
5. Else if quantity ≤ Reorder threshold, system flags item as **Low**, sends **Low-Stock Alert**, and **auto-triggers Module 20** — **System**
6. Else if quantity ≥ Overstock threshold, system flags item as **Overstock**, sends **Overstock Alert**, and **suppresses ordering** — **System**
7. System displays alerts on dashboard by severity — **System**
8. Doctor views alert; no approval needed for ordering — **Doctor**
9. Doctor reviews Overstock note and handles exceptions — **Doctor**
10. System logs alert and action in audit trail — **System**

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5. Outputs

- **Critical Low-Stock Alert** → Doctor (Module 21) + **Auto-Trigger** → **Module 20**

- **Low-Stock Alert** → Doctor (Module 21) + **Auto-Trigger** → **Module 20**
- **Overstock Alert** → Doctor (Module 21) for review; **ordering suppressed**
- **Flagged Item** → Module 20 (Critical/Low only)
- **Overstock Review Note** → Doctor (Module 21)
- **Alert/Action Audit Log** → System audit trail

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6. Stakeholders Analysis

Internal:

- **Doctor / Clinic Owner –** views alerts, reviews Overstock notes, handles exceptions
- **IT Department –** maintains alert logic and threshold configuration

External:

- **Suppliers** – receive orders automatically after Module 20 dispatch

Note: No receptionist is involved. Alerts are automatic, severity-based, and directly trigger ordering for Critical/Low.

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7. Flow Summary

Start → Monitor Stock → Compare to Critical Low, Low/Reorder, and Overstock Thresholds →

- Below Critical? → Yes → Critical Low-Stock Alert + **Auto-Trigger Module 20** → End
- No → Below Reorder? → Yes → Low-Stock Alert + **Auto-Trigger Module 20** → End
- No → Above Overstock? → Yes → Overstock Alert + **Suppress Ordering + Review Note** → End
- No → No Alert → End

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8. Performance Metrics

Metric	Current	Target
Stockout incidents	Common	Rare / 0
Critical stockout response	Not tracked	Immediate / same-day
Alert timing	During use – too late	In advance
Notification method	Manual	Automatic, severity-based
Ordering trigger	Manual	Automatic

Overstock incidents	Not tracked	Rare / 0
Overstock visibility	None	Real-time alert
Threshold accuracy	Manual / ad hoc	Reviewed periodically

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9. Issues and Challenges

- Low stock noticed only during use
- No advance warning
- No threshold monitoring
- No distinction between low and critical stock
- No overstock warning (excess inventory, tied-up capital, expiry risk)
- No link between pending orders and overstock
- Single-threshold logic may over-order or under-order
- Manual ordering creates delays

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10. Improvement Opportunities

- Multi-threshold monitoring: Critical Low, Low/Reorder, Overstock
- Severity-based auto-notification
- Automatic ordering trigger for Critical/Low
- Overstock suppression and review
- Dashboard visibility with color-coded alerts
- Integration with Module 19 (Expiration) and Module 20 (Ordering)

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11. Automation Potential

- Monitoring – **Automatic**
- Threshold comparison – **Automatic**
- Severity classification – **Automatic**
- Flagging – **Automatic**
- Notification – **Automatic**
- Ordering trigger – **Automatic**

Only remaining human touchpoint: Doctor views alerts, reviews Overstock notes, handles exceptions.

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12. Business Process Lifecycle Analysis

Phase	Description

Design	Set Critical Low, Reorder, and Overstock thresholds per item
Execution	System flags, notifies, and triggers Module 20 automatically
Monitoring	Track alert frequency, accuracy, stockout/overstock incidents
Analysis	Identify frequently flagged items and causes
Optimization	Adjust thresholds based on usage, lead time, seasonality, expiry

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13. Evaluation and Recommendation

Module 18 now gives advance warning for both shortages and excess, and directly triggers automatic ordering for Critical/Low conditions. **Critical alerts** trigger urgent replenishment, **low-stock alerts** trigger standard replenishment, and **overstock alerts** suppress ordering and prompt review.

Recommendations:

- Implement Module 18 after Module 17 – needs stock data.
- Set thresholds carefully: **Critical Low < Reorder < Overstock**.
- Use color-coded dashboard alerts for immediate visibility.
- Auto-trigger Module 20 for Critical/Low; suppress for Overstock.
- Review thresholds quarterly or after usage trends change.
- Integrate with Module 19 (Expiration) and Module 20 (Ordering).

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Module 19

PROCESS DOCUMENTATION

1. Information	
Process Name:	Expiration Date Monitoring Process
Process Owner:	Clinic Doctor / Clinic Owner
Department/Unit:	Inventory, Supplies & Operations
Document Prepared By:	Tristan Paolo C. De Castro
Date Prepared:	September 12, 2026
Version Number:	1.0
Last Updated:	September 12, 2026

2. Overview

Purpose: To automate the expiration-date checks the clinic already does manually.

Process Description: The system tracks expiration dates per supply batch, compares against the current date, and alerts the doctor a set number of days before expiry.

Actors Involved:
- System — tracks, compares, alerts
- Doctor — receives alert

3. Inputs

- Expiration Date per Batch - Supply entry
- Alert Window (days before expiry) - System config

4. Activities

1. System tracks expiration dates per batch - System
2. System compares against current date - System
3. System alerts doctor a set number of days before expiry - System
4. Alert appears on dashboard - System

5. Outputs

Expiration Alert -> Doctor (Module 21: Admin & Operations Dashboard)

6. Stakeholders Analysis

Internal:
- Doctor / Clinic Owner — receives alerts
- IT Department — maintains monitoring logic

External:
- None

Note: *No receptionist is involved. Alerts are automatic.*

7. Flow Summary

Start → Track Expiration Dates → Compare to Today → Near Expiry? → Yes → Alert Doctor → End

8. Performance Metrics	
Expiration checks	Manual, periodic (Target: Automatic, continuous)
Expired supplies used	Occasional (Target: 0%)
Alert timing	Late (Target: In advance)

9. Issues and Challenges

- Manual expiration checks are periodic
- Expired supplies may be used unknowingly
- No advance warning

10. Improvement Opportunities

- Auto-tracking
- Auto-alerts
- Dashboard visibility - with Module 21 (Admin & Operations Dashboard)

11. Automation Potential

- Tracking - Automatic
- Comparison - Automatic
- Alerting- Automatic

Only remaining human touchpoint: Doctor acts on alert.

12. Business Process Lifecycle Analysis	
Design	Define expiration alert windows.
Execution	System tracks and alerts automatically.
Monitoring	Track expired items and alert accuracy.
Analysis	Identify commonly expiring items.
Optimization	Adjust order quantities to reduce waste.

13. Evaluation and Recommendation

Module 19 automates what the clinic already does manually — with continuous monitoring instead of periodic checks.

Recommendation:

- Implement Module 19 after Module 17 (Supply Inventory Tracking) — needs supply data.
- Set an alert window at 30–60 days — enough time to use or reorder.
- Auto-flag on dashboard — immediate visibility

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Module 20

Module 20: Supply Ordering & Audit Process

1. Information

Field	Details
Process Name	Supply Ordering & Audit Process
Process Owner	Clinic Doctor / Clinic Owner
Department/Unit	Inventory, Supplies & Operations
Document Prepared By	Tristan Paolo C. De Castro
Date Prepared	September 12, 2026
Version Number	4.0
Last Updated	September 25, 2026

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2. Overview

Purpose: To fully automate supply ordering and auditing — removing the doctor's manual approval step — while respecting critical, low, and overstock thresholds. Orders are sent directly to suppliers, with a supplier message dispatched alongside the PO, and handed off to Module 22 for response tracking and automatic rerouting.

Process Description: When Module 18 flags **Critical Low** or **Low** stock, the system automatically drafts a restock order using supplier data from Module 23. At the **end of each day** (or immediately for critical items, if configured), the system generates a structured PDF Purchase Order **and** a supplier message (email/SMS), then dispatches both directly to the designated supplier. **No doctor approval is required.** Upon dispatch, the order is handed off to **Module 22 (Order Monitoring)** for response tracking. If the supplier replies **Unavailable** or **No Response**, Module 22 automatically reroutes the order to a backup supplier and logs the reroute **back into Module 20's audit trail** as a linked order. If Module 18 flags **Overstock**, Module 20 suppresses ordering entirely and logs a review note for the

doctor. All orders — including reroutes — are logged, and periodic audit reports are generated automatically.

Actors Involved:

- **System** – drafts orders, selects suppliers, sends POs + supplier messages, logs transactions, generates audits
- **Doctor** – receives notifications, reviews audit reports, handles exceptions only
- **Supplier** – receives and fulfills orders automatically
- **Module 22** – monitors supplier responses, reroutes on failure, returns reroute records to Module 20
- **Module 23** – provides supplier records, primary/backup chains, and routing data

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3. Inputs

- **Critical Low-Stock Alert** – Module 18
- **Low-Stock Alert** – Module 18
- **Overstock Alert** – Module 18 (for suppression only)
- **Live Stock Quantity** – Module 17
- **Expiration Data** – Module 19
- **Supplier Info** – Module 23: primary supplier, backup supplier chain, representative, email, phone, lead time, payment terms, response window
- **Order History** – Module 20
- **Reroute Records** – Module 22 (linked orders returned for audit)

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4. Activities

1. System receives threshold alert from Module 18 — **System**
2. System classifies alert as **Critical Low**, **Low**, or **Overstock** — **System**
3. If **Critical Low** or **Low**, system retrieves supplier mapping from Module 23 — **System**
4. System auto-drafts restock order with item, current stock, suggested quantity, primary supplier, and backup chain — **System**
5. System queues order for **end-of-day batch dispatch** (or immediate if critical and configured) — **System**
6. If **Overstock**, system suppresses auto-draft and logs a review note for the doctor — **System**
7. At scheduled dispatch time, system generates a structured PDF Purchase Order **and** a supplier message — **System**
8. System emails/SMS the PDF PO + supplier message directly to the designated supplier — **System**
9. System logs order (what, when, from whom, by whom) — **System**
10. System **hands off order to Module 22** for response tracking — **System**
11. Module 22 monitors the supplier reply and classifies it — **System / Module 22**
12. If Module 22 detects **Unavailable** or **No Response**, it reroutes to a backup supplier and **returns the reroute record to Module 20** — **Module 22 → System**
13. System logs the reroute as a **linked order** in the audit trail — **System**
14. System periodically generates an audit report — **System**
15. System feeds supplier performance data back to Module 23 — **System**
16. Doctor receives notification of orders sent, reroutes, and any exceptions — **Doctor**

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5. Outputs

- **Auto-Drafted Order** → System (internal)
- **Sent Purchase Order + Supplier Message** → Supplier
- **Order Confirmation Log** → Module 20
- **Reroute-Linked Order Record** → Module 20 audit trail (received from Module 22)
- **Supplier Audit Trail** → Module 20 / Doctor
- **Periodic Audit Report** → Doctor
- **Order Handoff Record** → Module 22 (for response tracking)
- **Supplier Performance Data** → Module 23
- **Overstock Review Note** → Doctor (Module 21)
- **Exception Alerts** (e.g., email bounce, reroute chain exhausted) → Doctor

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6. Stakeholders Analysis

Internal:

- **Doctor / Clinic Owner –** receives notifications, reviews audits, handles exceptions
- **IT Department –** maintains ordering system, supplier routing, and audit logic

External:

- **Suppliers (Primary) –** receive orders, reply with confirmation or unavailability
- **Suppliers (Backup) –** receive rerouted orders automatically when primary fails
- **Affiliate Companies –** linked suppliers under same parent or partner network

Note: No receptionist is involved. Ordering is fully automatic; the doctor is informed, not asked.

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7. Flow Summary

Start → Threshold Alert from Module 18 → Classify Alert → Critical Low or Low? → Yes → Retrieve Supplier + Backup Chain from Module 23 → Auto-Draft Order → Queue for End-of-Day Dispatch → Generate PDF PO + Supplier Message → Send to Supplier → Log Order → Hand Off to Module 22 → Module 22 Monitors Response → (Reroute if Unavailable / No Response) → Reroute Record Returned to Module 20 → Log Linked Order → Auto-Audit Report → Update Module 23 Performance → End

Alternative paths:

- **Overstock Alert** → No auto-draft → Log review note for doctor → End
- Supplier reply = **Unavailable** or **No Response** → Module 22 reroutes to backup → reroute logged in Module 20 → End
- Reroute chain exhausted → Escalate to Doctor → manual handling or **Cancelled**
- Email bounce on original dispatch → System alerts doctor for manual follow-up

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8. Performance Metrics

Metric	Current	Target
Ordering time	Manual, ad hoc	Fully automatic
Audit time	Hours, manual	Zero (auto)
Order accuracy	Human error	100%
Audit trail	None	Full, including reroutes
Critical alert response	Not tracked	Same-day dispatch
Supplier routing accuracy	Manual	Automatic via Module 23
Overstock suppression	None	Automatic
Reroute traceability	None	Full linked audit trail
Manual approvals required	Every order	Zero

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9. Issues and Challenges

- Ordering is reactive and manual
- Auditing supplies is time-consuming
- No audit trail
- No systematic reorder timing
- No distinction between critical and low stock in ordering
- No overstock control
- No centralized supplier data
- No order status visibility after submission
- Manual approval creates delays and bottlenecks
- No visibility on whether supplier accepted or rejected the order

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10. Improvement Opportunities

- Fully automatic order drafting and dispatch
- End-of-day batch processing to consolidate orders
- Immediate dispatch for critical items (configurable)
- Overstock suppression
- Full audit trail including reroute records
- Auto-generated audit report
- Supplier routing and backup chain via Module 23
- Response tracking and automatic rerouting via Module 22
- Supplier performance feedback to Module 23
- Dashboard visibility via Module 21

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11. Automation Potential

- Order drafting – **Automatic**
- Supplier selection – **Automatic** from Module 23
- Order dispatch – **Automatic** (end-of-day batch or immediate for critical)
- Supplier message dispatch – **Automatic**
- Response tracking – **Handed off to Module 22 (Automatic)**
- Reroute logging – **Automatic** (record received from Module 22)
- Audit report – **Automatic**

Only remaining human touchpoint: Doctor reviews audit reports and handles exceptions (email bounce, reroute chain exhausted, overstock review).

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12. Business Process Lifecycle Analysis

Phase	Description
Design	Create auto-drafting rules per threshold, supplier routing rules, backup chain rules, overstock suppression, end-of-day batch schedule, and audit report rules
Execution	System drafts, dispatches, logs, hands off to Module 22, and audits automatically
Monitoring	Track order frequency, response rates, reroute frequency, audit accuracy, supplier performance

Analysis	Identify supplier performance, reroute causes, and usage trends
Optimization	Adjust reorder points, thresholds, suppliers, backup chains, batch timing, and order quantities

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13. Evaluation and Recommendation

Module 20 now removes the doctor entirely from the ordering loop. Critical Low and Low alerts trigger automatic ordering, with dispatch batched at the end of each day. Module 23 ensures correct supplier routing and backup chains, Module 22 handles response monitoring and automatic rerouting, and reroute records flow back into Module 20's audit trail. The doctor is notified, not asked.

Recommendations:

- Implement **Module 23 (Supplier Management)** before Module 20 — Module 20 needs supplier data and backup chains.
- Implement **Module 18 (Low-Stock, Critical Low-Stock, and Overstock Alerts)** before Module 20 — Module 20 depends on threshold alerts.
- Implement Module 20 alongside **Module 22 (Order Monitoring)** — dispatch hands off directly to response tracking.
- Pre-load supplier info, including primary and backup chains, so drafts are ready to send.
- Configure end-of-day batch time (e.g., 6:00 PM) for non-critical orders.
- Optionally enable immediate dispatch for Critical Low items.
- Enable overstock suppression to prevent unnecessary purchasing.
- Auto-email audit reports monthly — zero effort.
- Review thresholds, supplier performance, and backup chains quarterly.
- Log all reroutes as linked orders so audits reflect the full order history.
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Module 22

Module 22: Order Monitoring Process

Revision Note: This version removes the **In Transit** status entirely and replaces it with a **Being Delivered** state that is set automatically once the supplier confirms. The system now **sends the order message to the supplier**, **parses the supplier's text response** to classify it as **Confirmed**, **Unavailable**, **No Response**, or **Unclear/Other**, and **automatically reroutes to a backup supplier** when the order cannot be fulfilled. The doctor's only manual action is clicking **Confirm Received** when supplies physically arrive.

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1. Information

Field	Details
Process Name	Order Monitoring Process
Process Owner	Clinic Doctor / Clinic Owner
Department/Unit	Inventory, Supplies & Operations
Document Prepared By	Tristan Paolo C. De Castro
Date Prepared	September 25, 2026
Version Number	3.0
Last Updated	September 25, 2026

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2. Overview

Purpose: To give the doctor a single, real-time view of every order generated by Module 20 — from automatic dispatch to physical receipt — while automatically handling supplier responses and rerouting to backup suppliers when needed, so no order silently fails.

Process Description: Module 20 dispatches the Purchase Order **and** a supplier message (email/SMS). Module 22 then tracks the order through a response-driven lifecycle. The system **reads the supplier's reply** and classifies it as **Confirmed**, **Unavailable**, **No Response** (timeout), or **Unclear**. Confirmed orders move automatically to **Being Delivered** — there is no separate "In Transit" status. When the supplier marks an item unavailable or fails to respond within the response window, the system **automatically reroutes the order to a backup supplier** via Module 23 and logs the reroute as a linked order. The doctor's only manual action is clicking **Confirm Received** once supplies arrive.

Actors Involved:

- **System** – sends order message, reads supplier response, classifies status, reroutes to backup, logs everything
- **Doctor** – views orders, classifies unclear responses, clicks **Confirm Received**, handles escalations
- **Supplier** – receives order, replies with confirmation or unavailability (external)
- **Module 23** – provides backup supplier data for automatic rerouting

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3. Order Status Model

Status	Set By	Meaning	Next Step
Queued for Dispatch	System	Order drafted, waiting for dispatch window	Sent
Sent / Awaiting Response	System	PO and message delivered to supplier; within response window	Awaits reply
Confirmed	System (parsed)	Supplier replied confirming the order	Auto → Being Delivered
Being Delivered	System	Order confirmed; awaiting physical arrival	Doctor confirms receipt

Unavailable	System (parsed)	Supplier replied that item(s) cannot be fulfilled	Auto → Reroute
No Response	System (timeout)	No reply within response window	Auto → Reroute
Unclear / Other	System (parsed)	Reply received but not classifiable	Doctor classifies manually
Rerouted	System	Order redirected to backup supplier; linked to new order	New sub-cycle begins
Received / Completed	Doctor	Doctor confirms supplies physically arrived	End
Cancelled	Doctor	Order terminated (no supplier available, or doctor decision)	End

Note: There is **no "In Transit" status**. Once the supplier confirms, the order is simply marked **Being Delivered** until the doctor confirms arrival.

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4. Inputs

- **Dispatched Order** – Module 20 (Supply Ordering & Audit)
- **Supplier Message / PO Email** – Module 20 (sent to supplier)
- **Supplier Reply** – External (email/SMS text)
- **Response Window** – Module 22 config (e.g., 24 hours; can be tied to supplier lead time)
- **Supplier Info & Backup Mapping** – Module 23 (Supplier Management)
- **Expected Lead Time** – Module 23
- **Order History** – Module 20

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5. Activities

1. System receives dispatched order from Module 20 — **System**
2. System assigns status **Sent / Awaiting Response** and starts the response timer — **System**
3. System monitors the supplier inbox for a reply — **System**
4. System parses the supplier's reply and classifies it — **System**
5. If reply = **Confirmed** → status becomes **Confirmed**, then auto-transitions to **Being Delivered — System**
6. If reply = **Unavailable** → status becomes **Unavailable**; system logs the reason — **System**
7. If no reply within the response window → status becomes **No Response**; system logs the timeout — **System**
8. If reply = **Unclear / Other** → system flags order for doctor classification — **System**
9. Doctor manually classifies unclear responses — **Doctor**
10. For **Unavailable** or **No Response**, system queries Module 23 for a backup supplier — **System**
11. System auto-creates a new order to the backup supplier and links it to the original order — **System**
12. Original order is marked **Rerouted**; new order enters its own **Sent / Awaiting Response** sub-cycle — **System**
13. If no backup supplier is available, or the reroute chain limit is reached, system escalates to the doctor — **System**
14. Doctor clicks **Confirm Received** when supplies physically arrive → status becomes **Received / Completed — Doctor**
15. System updates the order timeline and feeds performance data to Module 23 — **System**
16. Doctor views all orders, statuses, and reroutes on the monitoring dashboard — **Doctor**

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6. Outputs

- **Order Monitoring Dashboard** → Doctor (Module 21)
- **Order Status Timeline** → Doctor
- **Supplier Response Classification Log** → Module 20 / Module 23
- **Reroute Record (linked orders)** → Module 20 / Module 23
- **Escalation Alerts** → Doctor (Module 21)
- **Reroute Frequency Data** → Module 23 (Supplier Management)
- **Order History Log** → Module 20

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7. Flow Summary

Start → Module 20 Dispatches Order + Supplier Message → Status: Sent / Awaiting Response → System Reads Supplier Reply

- **Reply = Confirmed** → Confirmed → Auto: **Being Delivered** → Doctor Clicks **Confirm Received** → **Completed** → End

- **Reply = Unavailable** → Unavailable → Log → Query Module 23 for Backup → Auto-Order to Backup → Original = **Rerouted**, New Order begins sub-cycle → End
- **Reply = No Response (timeout)** → No Response → Log → Query Module 23 for Backup → Auto-Order to Backup → Original = **Rerouted** → End
- **Reply = Unclear / Other** → Flag for Doctor → Doctor Classifies → follows one of the above paths

Alternative paths:

- No backup supplier available → **Escalate to Doctor** → manual handling or **Cancelled**
- Backup supplier also fails to respond → next backup in chain (up to configured limit, e.g., 2) → then escalate
- Doctor clicks **Confirm Received** before status is Being Delivered → allowed, status jumps to **Completed**

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8. Response Classification Logic

Supplier Reply Pattern	Classification	System Action
"Confirmed," "Order received," "We'll process this," "Noted po"	**Confirmed**	→ Being Delivered
"Out of stock," "Unavailable," "Wala po kami stock," "Cannot fulfill"	**Unavailable**	→ Reroute to backup
No reply within response window	**No Response**	→ Reroute to backup
Partial reply (some items available, some not)	**Unclear / Other**	→ Doctor classifies; partial reroute may apply
Auto-reply, bounce, unrelated message	**Unclear / Other**	→ Doctor classifies
Question or clarification request from supplier	**Unclear / Other**	→ Doctor responds or classifies

Note: Response parsing uses keyword and pattern matching. Anything ambiguous is routed to the doctor rather than guessed — the system suggests, the doctor decides.

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9. Stakeholders Analysis

Internal:

- **Doctor / Clinic Owner** – views orders, classifies unclear responses, clicks **Confirm Received**, handles escalations
- **IT Department** – maintains parsing rules, response window config, and reroute logic

External:

- **Suppliers (Primary)** – receive orders, reply with confirmation or unavailability
- **Suppliers (Backup)** – receive rerouted orders automatically when primary fails

Note: No receptionist is involved. Supplier response handling and rerouting are automatic; the doctor only confirms physical arrival.

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10. Performance Metrics

Metric	Current	Target
Order visibility	Scattered / manual	Single dashboard
Status tracking	None	Real-time, response-driven
Supplier response capture	None	Automatic parsing
Reroute on failure	Manual	Automatic
Lost / failed orders	Common	Rare / 0
Doctor actions per order	Multiple	One (**Confirm Received**)

Overdue detection	Manual	Automatic timeout
Reroute traceability	None	Full linked audit trail

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11. Issues and Challenges

- No visibility on order status after submission to supplier
- No way to know if supplier received or accepted the order
- No tracking of delivery timelines
- No history of past orders
- Manual follow-up required for every order
- Risk of duplicate orders when status is unknown
- Supplier replies arrive in unstructured text (email/SMS), hard to classify
- Partial availability replies require human judgment

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12. Improvement Opportunities

- Automated supplier message dispatch (via Module 20)
- Automatic response parsing and classification
- **Automatic reroute to backup suppliers** on Unavailable or No Response
- Linked order audit trail (original → reroute)
- **Being Delivered** status replaces manual In Transit logging
- One-click **Confirm Received** for the doctor
- Escalation path when backup chain is exhausted
- Reroute frequency feeds Module 23 performance scoring

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13. Automation Potential

- Order dispatch – **Automatic** (Module 20)
- Response monitoring – **Automatic**
- Response classification – **Automatic** (with doctor fallback for unclear)
- Status updates – **Automatic**
- Backup rerouting – **Automatic**
- Reroute logging – **Automatic**
- Arrival confirmation – **Manual** (doctor clicks **Confirm Received**)

Only remaining human touchpoints: Doctor classifies unclear responses, clicks **Confirm Received**, and handles escalations.

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14. Business Process Lifecycle Analysis

Phase	Description
Design	Define response keywords, response window, backup chain limit, reroute rules
Execution	System sends, reads replies, classifies, reroutes automatically
Monitoring	Track response rates, reroute frequency, confirmation times
Analysis	Identify suppliers that frequently fail or go silent
Optimization	Reassign suppliers, adjust response windows, update backup lists

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15. Evaluation and Recommendation

Module 22 now runs on **supplier responses rather than assumed logistics**. It removes the unreliable "In Transit" status, replaces it with an automatic **Being Delivered** state, and gives the doctor a single action — **Confirm Received**. Failed or unanswered orders are **automatically rerouted to backup suppliers** and fully logged, so no order silently dies. Reroute data feeds directly into Module 23's supplier performance scoring, closing the loop.

Recommendations:

- Implement Module 22 alongside **Module 20 (Supply Ordering & Audit)**.
- Implement **Module 23 (Supplier Management)** first — rerouting depends on backup supplier mapping.
- Configure the **response window** (e.g., 24 hours) per supplier or globally.
- Set a **reroute chain limit** (e.g., maximum 2 backups) to prevent infinite loops.
- Route all ambiguous replies to the doctor — never guess.
- Surface reroute events and escalations prominently on the Module 21 dashboard.
- Feed reroute frequency into Module 23 supplier scores.
- Allow the doctor to override classification manually at any time, with an audit log entry.
-

Module 23

Module 23: Supplier Management Process

1. Information

Field	Details
Process Name	Supplier Management Process
Process Owner	Clinic Doctor / Clinic Owner
Department/Unit	Inventory, Supplies & Operations
Document Prepared By	Tristan Paolo C. De Castro
Date Prepared	September 25, 2026
Version Number	1.2
Last Updated	September 25, 2026

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2. Overview

Purpose: To centralize all supplier information — affiliate companies, representatives, contact details, product mappings, and **ordered backup supplier chains** — so ordering is fast, reliable, and resilient, with automatic routing to the next available backup supplier when a primary supplier is unavailable or unresponsive.

Process Description: The system maintains a master supplier database. Each supplier record includes company name, contact person/representative, phone, email, address, products supplied, lead time, **response window**, payment terms, and status (Active / Inactive / Backup).

Primary suppliers and an **ordered backup chain** are linked to specific inventory items so Module 20 can auto-route orders and Module 22 can auto-reroute when the primary fails. When a primary supplier replies **Unavailable** or **No Response**, Module 22 moves down the backup chain automatically, logging every reroute back into Module 20 and into Module 23 as a performance event.

Actors Involved:

- **Doctor** – adds, edits, views supplier records; defines primary and backup chain; reviews performance
- **System** – stores, links, retrieves supplier data; supplies backup chains to Module 22 for rerouting; tracks performance
- **Suppliers** – external parties whose details are recorded

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3. Inputs

- **Supplier Information** – Doctor (manual entry): company name, representative, contact details, address
- **Product–Supplier Mapping** – Doctor (which items each supplier provides)
- **Primary Supplier + Ordered Backup Chain** – Doctor
- **Lead Time, Response Window & Payment Terms** – Doctor
- **Order History** – Module 20 (Supply Ordering & Audit)
- **Supplier Performance Data** – Module 22 (Order Monitoring): response rates, response times, reroute frequency, delivery confirmation

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4. Activities

1. Doctor adds new supplier record — **Doctor**
2. System stores company name, representative, contact details, products, lead time, response window, payment terms — **System**
3. Doctor links suppliers to inventory items, defining a **primary supplier and an ordered backup chain — Doctor**
4. System uses supplier data to auto-draft and route orders in Module 20 — **System**
5. When the primary supplier replies **Unavailable** or **No Response**, Module 22 queries the system for the **next supplier in the backup chain — Module 22 / System**
6. System provides the next backup supplier automatically — **System**
7. Reroute event is logged back into Module 23 as a supplier performance event — **System**
8. System tracks supplier performance from order history and reroute records — **System**
9. Doctor views supplier list, backup chains, and performance on dashboard — **Doctor**
10. Doctor updates supplier details, response windows, and backup chain order as needed — **Doctor**
11. System logs all supplier changes in audit trail — **System**

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5. Outputs

- **Supplier Master List** → Doctor
- **Supplier–Item Mapping with Ordered Backup Chain** → Module 20 (routing) and Module 22 (rerouting)
- **Supplier Performance Report** → Doctor (Module 21)
- **Reroute Frequency Data** → Doctor (Module 21) and Module 22
- **Supplier Contact Directory** → Doctor
- **Order Routing Rules** → Module 20
- **Backup Chain Configuration** → Module 22

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6. Stakeholders Analysis

Internal:

- **Doctor / Clinic Owner** – manages supplier records, defines backup chains, reviews performance
- **IT Department** – maintains supplier database, routing logic, and backup chain resolution

External:

- **Suppliers (Primary)** – receive orders, reply with confirmation or unavailability
- **Suppliers (Backup)** – receive rerouted orders automatically when earlier suppliers in the chain fail
- **Affiliate Companies** – linked suppliers under same parent or partner network

Note: No receptionist is involved. Supplier data is centralized and maintained by the doctor. Backup routing follows the configured chain automatically.

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7. Flow Summary

Start → Doctor Adds Supplier → System Stores Record → Doctor Links to Items with Primary + Backup Chain → Module 20 Uses Supplier Data → Orders Routed → Module 22 Monitors Response → If Primary Fails → System Provides Next Backup in Chain → Reroute Logged → Performance Tracked → Doctor Reviews → End

Alternative paths:

- Primary supplier confirms → no reroute → order proceeds to Being Delivered
- Primary replies **Unavailable** → Module 22 reroutes to **Backup #1** → logged
- Primary gives **No Response** (timeout) → Module 22 reroutes to **Backup #1** → logged
- Backup #1 also fails → move to **Backup #2** (up to configured chain limit)
- Entire chain exhausted → Escalate to Doctor → manual handling or **Cancelled**
- Supplier performance poor → Doctor reassigns items or reorders the backup chain

- New supplier added → Record created, linked to items, available for ordering and backup chains

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8. Performance Metrics

Metric	Current	Target
Supplier info	Scattered / manual	Centralized
Contact lookup time	Minutes	Seconds
Backup supplier readiness	None	Ordered chain, pre-linked
Order routing accuracy	Manual	Automatic
Supplier performance visibility	None	Full history
Supplier data maintenance	Ad hoc	Structured, audited
Response Rate	Not tracked	High (% of orders replied to)
Average Response Time	Not tracked	Within configured window
Reroute Frequency	Not tracked	Low (few reroutes per supplier)
Backup Chain Depth Used	Not tracked	Rarely beyond Backup #1

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9. Issues and Challenges

- Supplier information scattered across phone contacts, messages, and paper
- No backup supplier list
- No performance tracking
- Manual lookup during ordering
- No centralized contact directory
- No way to compare supplier reliability or pricing
- Manual backup activation causes delays
- No visibility on which suppliers go silent or frequently reply Unavailable
- No structured backup chain — single backup only was previously assumed

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10. Improvement Opportunities

- Centralized supplier database
- **Ordered backup supplier chain per item** (Primary → Backup #1 → Backup #2 …)
- Auto-routing of orders to correct supplier
- Automatic rerouting to next backup in chain via Module 22
- Supplier performance tracking from Module 22, including **Response Rate**, **Response Time**, and **Reroute Frequency**
- One-click supplier contact and order history
- Affiliate company grouping for bulk negotiations
- Response window configuration per supplier

---

11. Automation Potential

- Storage – **Automatic**
- Retrieval – **Automatic**
- Order routing – **Automatic**
- Backup chain resolution – **Automatic**
- Reroute frequency tracking – **Automatic**
- Performance tracking – **Automatic**
- Contact directory – **Automatic**

Only remaining human touchpoint: Doctor adds and updates supplier records, configures backup chains, and reviews performance.

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12. Business Process Lifecycle Analysis

Phase	Description

Design	Create supplier master list with primary + ordered backup chain per item; define response windows
Execution	System stores, links, routes, and resolves backup chains automatically
Monitoring	Track supplier response rate, response time, reroute frequency, and delivery confirmation
Analysis	Identify reliable vs. unreliable suppliers and frequent chain failures
Optimization	Reassign items, reorder backup chains, adjust response windows, negotiate terms

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13. Evaluation and Recommendation

Module 23 eliminates the need to search for supplier contacts and ensures orders always have a destination — even when a primary supplier replies **Unavailable** or goes silent. The **ordered backup chain** gives Module 22 a deterministic reroute path, and every reroute is logged as a performance event, so the doctor can see which suppliers fail most often.

Recommendations:

- Implement Module 23 before Module 20 — Module 20 needs supplier data and backup chains to auto-draft and route orders.
- Pre-load all suppliers, including backups and affiliate companies.
- Link every inventory item to a **primary supplier and an ordered backup chain** (at least Backup #1; ideally Backup #2).
- Configure **response windows** per supplier (e.g., 24 hours) — this drives Module 22's timeout logic.
- Set a **chain limit** (e.g., maximum 2 backups) to prevent infinite loops; escalate to the doctor when exhausted.
- Track **Response Rate**, **Response Time**, and **Reroute Frequency** automatically using Module 22 data.
- Review supplier list, backup chain order, and terms quarterly.
- Group affiliate companies for potential bulk ordering or shared terms.

Module 21

Module 21: Doctor's Daily Workflow Process

1. Information

Field	Details
Process Name	Doctor's Daily Workflow Process
Process Owner	Clinic Doctor / Clinic Owner
Department/Unit	Admin & Operations Dashboard
Document Prepared By	Tristan Paolo C. De Castro
Date Prepared	September 12, 2026
Version Number	2.1
Last Updated	September 25, 2026

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2. Overview

Purpose: To give the doctor one control-center view tying every other module together — now including automatic order notifications, supplier response statuses, reroute events, escalations, and a one-click **Confirm Received** button — directly addressing the daily-repeated-task and pain-point questions from the interview.

Process Description: The doctor logs into the dashboard and sees today's appointments, pending recalls, overdue balances, low-stock items, automatically sent orders, supplier response statuses, reroute events, escalations, and daily income in one screen. Since ordering is now fully automatic and Module 22 handles supplier responses and rerouting, the doctor no longer approves restock orders — they review notifications, monitor response-driven order statuses, confirm physical arrival with one click, and handle only exceptions (unclear supplier replies, reroute chain exhaustion, overstock reviews).

Actors Involved:

- **Doctor** – views dashboard, confirms received supplies, classifies unclear responses, handles exceptions, takes one-click actions
- **System** – aggregates data from all modules

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3. Inputs

- **Today's Appointments** – Module 4 (Schedule & Conflict Management)
- **Pending Recalls** – Module 5 (Patient Registration & Authentication)
- **Overdue Balances** – Module 15 (Outstanding Balance Monitoring)
- **Low-Stock / Critical / Overstock Alerts** – Module 18
- **Sent Orders / Order Notifications** – Module 20 (Supply Ordering & Audit)
- **Supplier Response Status** – Module 22 (Order Monitoring): Confirmed, Unavailable, No Response, Unclear
- **Being Delivered Orders (Pending Confirm Received)** – Module 22
- **Reroute Events** – Module 22 (linked order history)
- **Escalations** – Module 22 (reroute chain exhausted, ambiguous replies)
- **Overstock Review Notes** – Module 20
- **Daily Income** – Module 16 (Income & Expense Reporting)

---

4. Activities

1. Doctor logs into dashboard — **Doctor**
2. Dashboard shows all key metrics — **System**
3. Doctor reviews flagged items (Critical/Low stock, Overstock notes) — **Doctor**
4. Doctor views supplier response statuses (Confirmed, Unavailable, No Response, Unclear) — **Doctor**
5. Doctor views reroute events and linked order history — **Doctor**
6. Doctor classifies any unclear supplier responses — **Doctor**
7. Doctor clicks **Confirm Received** for orders in **Being Delivered** status once supplies physically arrive — **Doctor**
8. Doctor handles escalations (reroute chain exhausted, email bounce) — **Doctor**
9. Doctor assigns walk-in slots (one-click) — **Doctor**
10. Doctor charts + taps supplies used (during visits) — **Doctor**
11. Doctor views end-of-day income summary — **Doctor**

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5. Outputs

- **Completed Daily Workflow** → Doctor
- **Confirm Received Actions** → Module 22 → Module 20 → Module 17 (stock updated)
- **Response Classifications** → Module 22
- **Exception Handling** → Module 20 / Module 22
- **Overstock Review** → Module 20
- **Walk-in Assignments** → Module 4
- **Chart Entries** → Module 9 & Module 17

- **Daily Income View** → Doctor

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6. Stakeholders Analysis

Internal:

- **Doctor / Clinic Owner** – primary user
- **IT Department** – maintains dashboard

External:

- None directly (patients benefit indirectly; suppliers interact via Module 20/22 automation)

Note: No receptionist is involved. Dashboard replaces the receptionist's coordination role and removes the doctor's manual approval duties. Supplier response handling is automatic; the doctor only confirms receipt and handles exceptions.

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7. Flow Summary

Start → Log In → View Dashboard → Review Flagged Items / Sent Orders / Supplier Response Statuses / Reroute Events → Classify Unclear Responses (if any) → Click Confirm Received (for Being Delivered orders) → Handle Escalations (if any) → One-Click Actions (exceptions only) → Chart During Visits → View Income → End

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8. Performance Metrics

Metric	Current	Target
Non-clinical time per day	1–2 hours	Under 15 minutes
Dashboard views	Multiple (calendar, chart, supplies)	Single screen
Manual data entry	Frequent	Zero
Manual order approvals	Every order	Zero

Income visibility	End of day, manual	Real-time
Supplier response visibility	None	Real-time
Reroute visibility	None	Real-time, linked history
Doctor actions per order	Multiple	One (**Confirm Received**)

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9. Issues and Challenges

- Doctor checks calendar, chart shelf, supply list separately
- Manual income computation
- No unified view
- Constant context-switching
- Previously required manual order approvals (now removed)
- Previously no visibility on supplier responses or reroutes (now surfaced)

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10. Improvement Opportunities

- Single dashboard
- One-click actions (exceptions only)
- Real-time metrics
- Zero data entry
- Automatic order notifications
- Supplier response status visibility
- Reroute event and linked order history visibility
- One-click **Confirm Received** for physical arrival

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11. Automation Potential

- Data aggregation – **Automatic** from all modules
- Order dispatch – **Automatic** (Module 20)
- Supplier response tracking – **Automatic** (Module 22)
- Reroute to backup – **Automatic** (Module 22)
- Reporting – **Automatic**

Only remaining human touchpoints: Doctor classifies unclear responses, clicks **Confirm Received**, and handles escalations.

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12. Business Process Lifecycle Analysis

Phase	Description
Design	Create a single dashboard aggregating all modules, including response and reroute widgets
Execution	Doctor views, confirms receipt, classifies exceptions, and handles escalations
Monitoring	Track dashboard usage, response handling time, confirmation lag
Analysis	Identify most-used features and pain points
Optimization	Reorder/redesign based on usage

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13. Evaluation and Recommendation

Module 21 remains the single biggest time-saver for the doctor — it replaces 4–5 separate manual checks with one screen and one-click actions. With automatic ordering and response-driven tracking now in place, the doctor's role is reduced to **reviewing, confirming receipt, classifying unclear responses, and handling escalations**.

Recommendations:

- Implement Module 21 last — needs data from all other modules.
- Design for one-screen visibility — no scrolling.
- Prioritize one-click actions — every approval should be one tap.
- Display supplier response statuses and reroute events prominently.
- Surface only exceptions (unclear replies, reroute exhaustion, bounces, overstock) as actionable items.
- Provide a dedicated **Confirm Received** button next to every **Being Delivered** order.
- Show linked order history for reroutes so the doctor can trace the full chain.
-